Logical operators example truth table

Input			Output	
p	q	r	$(p \wedge q) \oplus (\ (p \oplus q) \wedge r\ )$	$(p \wedge q) \vee (\ (p \oplus q) \wedge r\ )$
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

Logical equivalence

Logical equivalence : Two compound propositions are **logically equivalent** means that they have the same truth values for all settings of truth values to their propositional variables.

Tautology: A compound proposition that evaluates to true for all settings of truth values to its propositional variables; it is abbreviated T.

Contradiction: A compound proposition that evaluates to false for all settings of truth values to its propositional variables; it is abbreviated F.

Contingency: A compound proposition that is neither a tautology nor a contradiction.

Logical equivalence extra example

Extra Example: Which of the compound propositions in the table below are logically equivalent?

Input		Output				
p	q	$\neg(p \wedge \neg q)$	$\neg(\neg p \vee \neg q)$	$(\neg p \vee q)$	$(\neg q \vee \neg p)$	$(p \wedge q)$
T	T					
T	F					
F	T					
F	F					

CC BY-NC-SA 2.0 Version October 15, 2026 (1)

Logical equivalence identities andornotxor

(Some) logical equivalences

Can replace p and q with any compound proposition

$\neg(\neg p) \equiv p$				**Double negation**
$p \vee q \equiv q \vee p$		$p \wedge q \equiv q \wedge p$		**Commutativity** Ordering of terms
$(p \vee q) \vee r \equiv p \vee (q \vee r)$		$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$		**Associativity** Grouping of terms
$p \wedge F \equiv F$	$p \vee T \equiv T$	$p \wedge T \equiv p$	$p \vee F \equiv p$	**Domination** aka short circuit evaluation
$\neg(p \wedge q) \equiv \neg p \vee \neg q$		$\neg(p \vee q) \equiv \neg p \wedge \neg q$		**DeMorgan's Laws**

CC BY-NC-SA 2.0 Version October 15, 2026 (2)